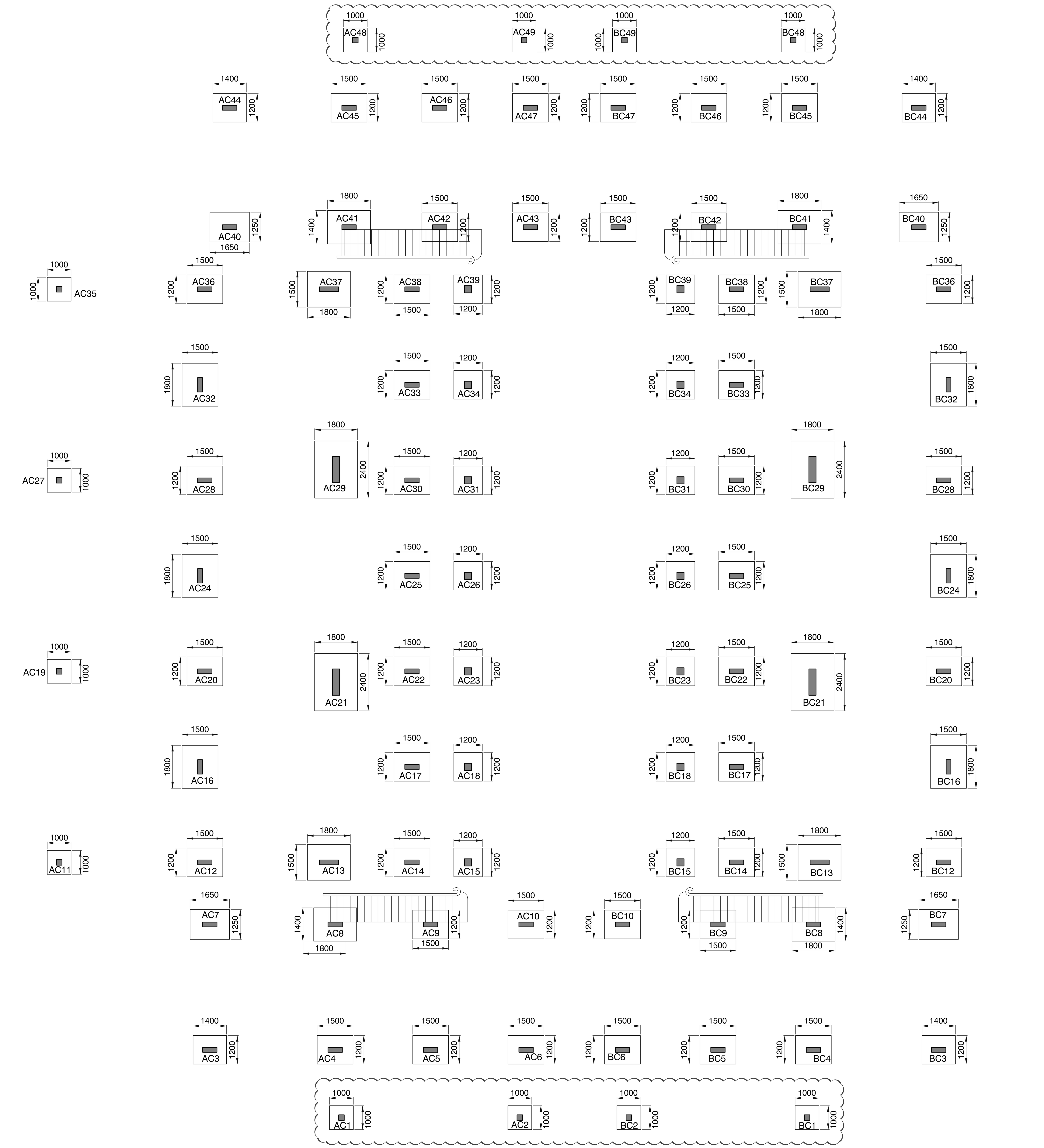
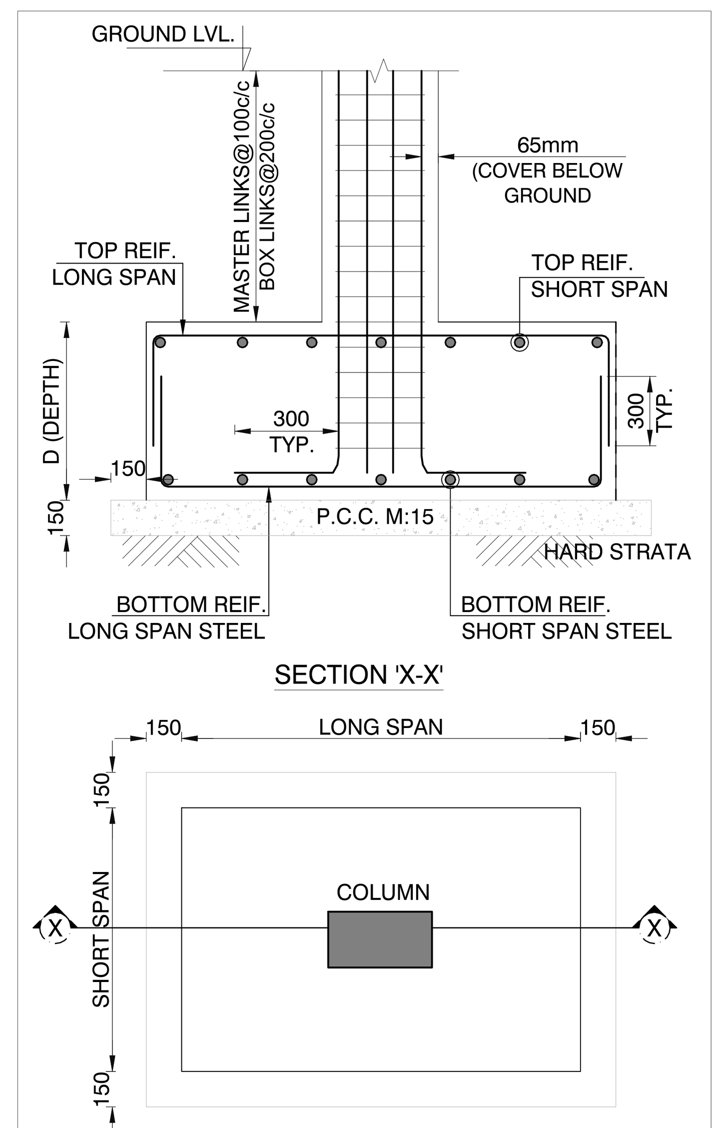
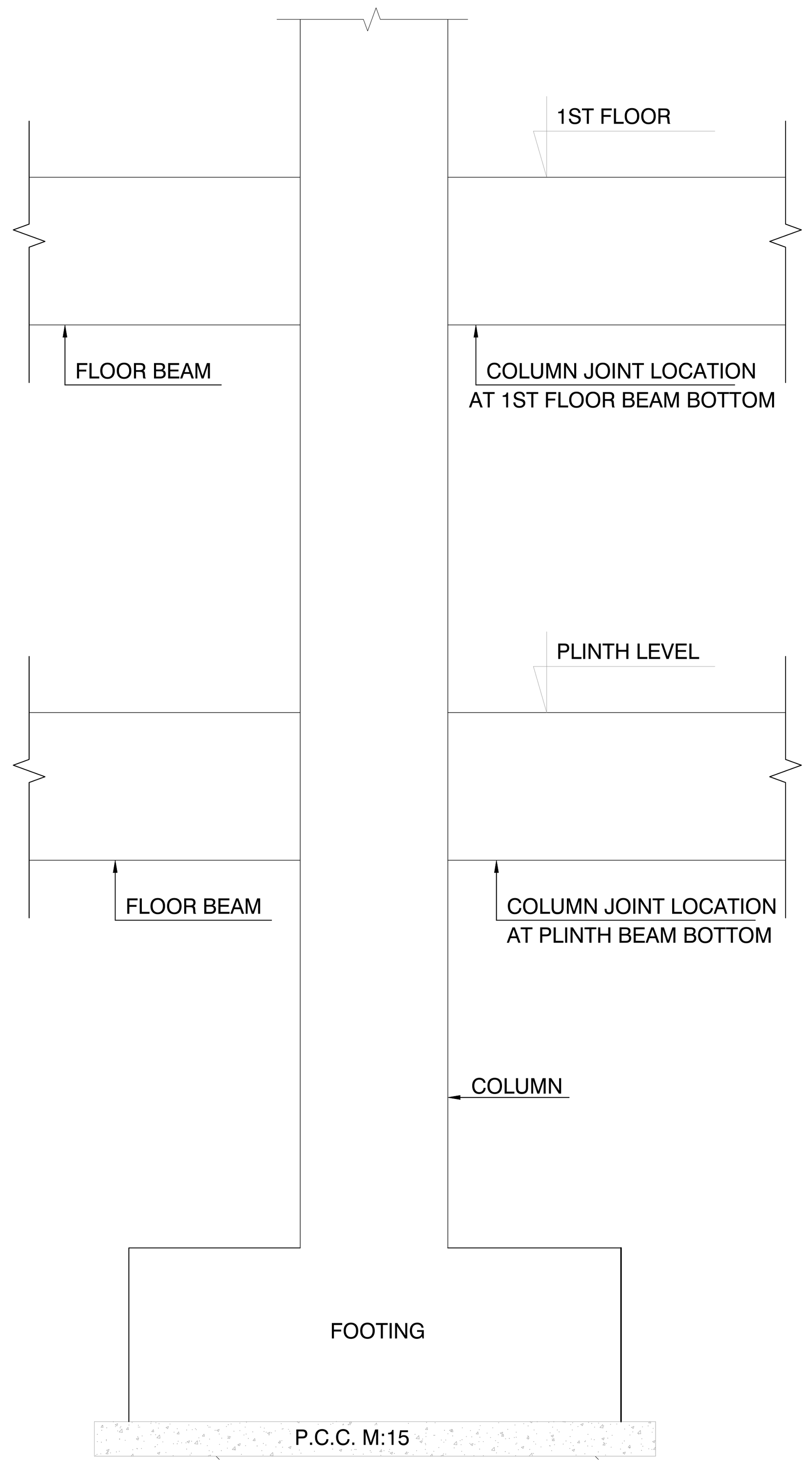


FOUNDATION LAYOUT



TYPICAL DETAILS OF FOOTING



TYPICAL DETAIL FOR COLUMN JOINT LOCATION

Development length of bar in Tension (Slab, Beam & Footing)

TYPE OF STEEL	Value of K for concrete mixes				
	M20	M25	M30	M35	M40 & ABOVE
Fe 415	47 x D	40 x D	38 x D	33 x D	30 x D
Fe 500	57 x D	49 x D	45 x D	40 x D	36 x D

Development length of bar in Compression (Column)

TYPE OF STEEL	Value of K for concrete mixes				
	M20	M25	M30	M35	M40 & ABOVE
Fe 415	38 x D	32 x D	30 x D	30 x D	30 x D
Fe 500	46 x D	39 x D	36 x D	32 x D	30 x D

COLUMN SCHEDULE

FIRST FLOOR TO ROOF LEVEL	M25	SIZE	- - - -	200 x 600	200 x 600	200 x 600
		STEEL	- - - -	8-T20 + 2-T16	10-T20	8-T20 + 4-T16
GROUND FLOOR TO FIRST FLOOR	M25	SIZE	- - - -	200 x 600	200 x 600	200 x 600
		STEEL	- - - -	8-T20 + 2-T16	4-T25 + 6-T20	8-T20 + 4-T16
FOOTING TO GROUND FLOOR	M25	SIZE	230 x 230	200 x 600	200 x 600	200 x 600
		STEEL	4-T16	8-T20 + 2-T16	4-T25 + 6-T20	8-T20 + 4-T16
LINK (Refer Typ. Detail for Spacing)			T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)
COLUMN MARKED			**AC1,2,48,49**	**AC3,5,6,9,10,42, AC43,44,46,47**	**AC4,8,41,45**	**AC7,40**
			BC1,2,48,49	**BC3,5,6,9,10,42, BC43,44,46,47**	**BC4,8,41,45**	**BC7,40**

COLUMN SCHEDULE

FIRST FLOOR TO ROOF LEVEL	M25	SIZE	- - - -	- - - -	- - - -	- - - -	- - - -	- - - -
		STEEL	- - - -	- - - -	- - - -	- - - -	- - - -	- - - -
GROUND FLOOR TO FIRST FLOOR	M25	SIZE	200 x 200	200 x 600	200 x 800	230 x 230	200 x 600	200 x 1100
		STEEL	4-T16	4-T25 + 6-T20	4-T25 + 12-T20	4-T16 + 4-T12	4-T25 + 8-T20	4-T25 + 16-T20
FOOTING TO GROUND FLOOR	M25	SIZE	230 x 230	200 x 600	200 x 800	300 x 300	200 x 600	300 x 1100
		STEEL	4-T16	4-T25 + 6-T20	8-T25 + 8-T20	4-T16 + 6-T12	4-T25 + 8-T20	8-T25 + 14-T20
LINK (Refer Typ. Detail for Spacing)			T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)	T8 @100C/C (Master) +T8 @200C/C (Box)
COLUMN MARKED			**AC11,19,27,35**	**AC12,14,17,22, AC25,30,33,38**	**AC13,37**	**AC15,18,23,26, AC31,34,39**	**AC16,20,24,28, AC32,36**	**AC21,29**
			BC11,19,27,35	**BC12,14,17,22, BC25,30,33,38**	**BC13,37**	**BC15,18,23,26, BC31,34,39**	**BC16,20,24,28, BC32,36**	**BC21,29**

FOOTING SCHEDULE

TOP STEEL	M25 : Fe500	SHORT DIR	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -
		LONG DIR	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -
BOTTOM STEEL		SHORT DIR	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T12@100 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T12@100 C/C
		LONG DIR	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T12@100 C/C	T10@150 C/C	T10@150 C/C	T10@150 C/C	T12@100 C/C
DETAIL OF FOOTING		DEPTH (D)	400	450	450	450	450	450	450	750	600	450	600	800
		SIZE	1000 x 1000	1200 x 1400	1200 x 1500	1250 x 1650	1400 x 1800	1200 x 1500	1200 x 1500	1500 x 1800	1200 x 1500	1200 x 1200	1500 x 1800	1800 x 2400
P.C.C.	M15	150 THK.	1300 x 1300	1500 x 1700	1500 x 1800	1550 x 1950	1700 x 2100	1500 x 1800	1500 x 1800	1800 x 2100	1500 x 1800	1500 x 1500	1800 x 2100	2100 x 2700
COLUMNS MARKED			**AC1,2,11,19,27,35 AC48,49**	**AC3,44**	**AC4,5,6, AC45,46,47**	**AC7,40**	**AC8,41**	**AC9,10,42,43**	**AC12,20,28,36**	**AC13,37**	**AC14,17,22,25, AC30,33,38**	**AC15,18,23,26, AC31,34,39**	**AC16,24,32**	**AC21,29**
			BC1,2,11,19,27,35 BC48,49	**BC3,44**	**BC4,5,6, BC45,46,47**	**BC7,40**	**BC8,41**	**BC9,10,42,43**	**BC12,20,28,36**	**BC13,37**	**BC14,17,22,25, BC30,33,38**	**BC15,18,23,26, BC31,34,39**	**BC16,24,32**	**BC21,29**

DRAWING NOS. - ASC - 02

PROJECT STAGE :

☐ - SUB. ☐ - TENDER ☐ - ADVANCE ☑ - **EXECUTION**

SPECIAL NOTES :-

ALL EXTERNAL WALLS SHALL BE 150 thk. LIGHT WEIGHT BLOCKS.
ALL INTERNAL WALLS SHALL BE 150 thk. LIGHT WEIGHT BLOCKS.

NOTES :

1. Basic reference code:- IS 456: 2000, IS 1893, IS 13920
2. Due care shall be taken to ascertain that requisite strength of concrete is gained before commencement of deshuttering. It shall comply with provisions of Clause No. 11.3 of IS 456: 2000.
3. Earthquake zone considered is zone III for MUMBAI region.
4. Clear covers
   - a) Footings / Raft (Bottom / Top) 75/50
   - b) Columns & walls >200mm width 40
   - c) Columns & walls having width of 200mm & below 25
   - d) Slabs 20
   - e) Beams 25
   - f) Girder Beams 40
   - g) Pile cap 75
5. Beams having depth more than 750mm, provide side-face reinforcement.
6. Substratum shall be approved from our office before laying P.C.C.
7. Minimum clear spacing between any two longitudinal bars in beam= 50mm.
8. All laps (Ld) shall be staggered & not more than 50% bars to be lapped at any given section. Ld = 50 X D (where D= dia. of Bar)
9. All buildings shall have tie beams/plinth beams at ground/plinth level.
10. At any level where column size gets reduced in either dimension tie beams/plinth beams are essential.
11. For cantilevers, top bars to be anchored behind from external face of support for -1.5 X span of cantilever.
12. Fire rating considered:- 2 Hour Max.

Use of this drawing for construction shall explicitly confirm acceptance of following conditions by Owner / Builder / Contractor

1. Our responsibility shall remain limited to safe and sound structural design as transmitted by this drawing and we shall not remain responsible for
   - a) Safety of old structure during demolition.
   - b) Safety of any adjoining building /persons staying in adjoining building/persons and properties on adjoining roads.
   - c) Safety of construction worker/any personnel at work site during construction
   - d) Correctness/safety of any temporary structure, scaffolding, shuttering, centering erected at site and any injury to any personnel arising out of any accidents.
   - e) Accidents occurring due to premature deshuttering, faulty / substandard construction material or workmanship / faulty construction procedure.
   - f) Any accident occurring due to construction of elements of buildings not designed by us.
2. Supervision if specifically asked for will be provided to the extent of verification of reinforcement on site
3. All structural concrete should be weigh batched, machine mixed & mechanically vibrated.
4. Any discrepancy between our drawing & Architects' drawing shall be brought to our notice before construction.
5. Drawing shall not be scaled written dimensions are to be followed.
6. This drawing to be read along with other relevant drawings.

REV.	DESCRIPTION	DATE	DWN.	CHKD.
R9	REVISED AS MARKED	07.03.24	NK	SIDDHI
R8	REVISED AS PER COMMENT & AS MARKED	13.10.23	NK	SIDDHI
R7	REVISED AS MARKED	10.10.23	NK	SIDDHI
R6	REVISRD A PER COMMENTS	15.09.23	NK	SIDDHI
R5	NEW COLUMNS ARE DELETED	13.09.23	NK	SIDDHI
R4	NEW COLUMNS ADDED REVISED AS MARKED	12.09.23	NK	SIDDHI
R3	REVISED AS PER CENTERLINE PLAN	05.09.23	NK	ROHIT
R2	REVISED AS MARKED	04.09.23	NK	SIDDHI
R1	REVISED AS PER LATEST ARCH. DRAWING	31.08.23	NK	ROHIT

GRADE OF CONCRETE :-	- M25
GRADE OF STEEL :-	- Fe500
ENVIRONMENTAL EXPOSURE CONDITION :-	- MODERATE
DESIGN LIVE LOAD :- (UNLESS SPECIFIED)	-
S.B.C. :-	- 35 T/m²

PROJECT : PROPOSED RESORT @ PALI, RAIGARH
(RESIDENTIAL BLOCK - 1A & 1B)

CLIENT :	MELLORA INFRASTRUCTURE
ARCHITECT :	STUDIO HUMANE
DRAWING TITLE :	RCC DETAIL OF FOOTING,COLUMN SCHEDULE, COLUMN LINKS & GENERAL DETAILS

SCALE	DWN.BY	CHK.BY	DATE	JOB NO.
N.T.S.	VASANT	NAMRATA	21.04.2023	ASC-2544

Associated Structural Engineers LLP.
607, Opal Square, Opp. Railadevi Lake, Barve Rd. Wagale Estate, Thane (w)- 400 604.
Phone: +91-22-20812084/85, 8976072223
Email: info@asc.org.in Website: www.asc.org.in